Replication of Card & Krueger (1994): Minimum Wages and Employment

A Difference-in-Differences Analysis of the New Jersey Fast-Food Industry

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1. Research question and motivation

Question. Did New Jersey's April 1, 1992 increase in the minimum wage from $4.25 to $5.05 per hour reduce employment in fast-food restaurants?

Motivation. The paper is the canonical application of the difference-in-differences (DiD) design and the most influential empirical test of the competitive theory of the labour market. A textbook competitive model predicts that a binding minimum wage reduces employment along the labour-demand curve. Card and Krueger exploit a natural experiment: New Jersey (NJ) raised its minimum wage while neighbouring eastern Pennsylvania (PA), across the Delaware River, left its wage floor unchanged at $4.25. The two states share a regional labour market, making PA a credible comparison group.

Hypotheses. Under the competitive benchmark the NJ–PA difference in employment growth should be negative. Under monopsony or search-friction models, employment may be unchanged or even rise. The paper's headline finding — no evidence of an employment decline, and if anything a relative increase — is the result we set out to reproduce.

2. Data sources, sample construction, and variable definitions

Source. Public replication data from Card & Krueger (1994), distributed as the Princeton restaurant survey (`public.csv`). The file contains 410 store-level observations (46 columns): 331 stores in New Jersey (`STATE = 1`) and 79 in eastern Pennsylvania (`STATE = 0`), from four chains — Burger King (171), KFC (80), Roy Rogers (99), and Wendy's (60). Stores were surveyed twice: wave 1 in February–March 1992 (before the increase) and wave 2 in November–December 1992 (after).

Sample construction. The core analysis uses a balanced panel of stores that answered wave 2 and remained open (`STATUS2 = 1`): 399 stores (321 NJ, 78 PA). Eleven stores are excluded: 6 permanent closures, 2 renovation closures, 1 highway-construction closure, 1 mall-fire closure, and 1 refusal. After listwise deletion of observations missing employment components in either

wave, the FTE regressions use 378 stores. We do not publish the raw data; see `README.md` for access instructions.

Variable definitions.

Variable	Definition
FTE employment	`EMPFT + 0.5 * EMPPT + NMGRS` (full-time + managers + half of part-time)
Headcount	`EMPFT + EMPPT + NMGRS`
`NJ` (treatment)	1 if `STATE == 1`, 0 if Pennsylvania
Change in FTE, `dfte`	wave-2 FTE minus wave-1 FTE
Menu prices	`PSODA`, `PFRY`, `PENTREE` (medium soda, small fries, entree, incl. tax)
Controls	chain dummies (`CHAIN`), company-owned (`CO_OWNED`), NJ sub-region dummies (`SOUTHJ`, `CENTRALJ`, `SHORE`; north NJ omitted), initial wage `WAGE_ST`

Cleaning log. All cleaning is coded in `src/prep.py` and reproduced by `run_all.py`. Missing starting wages are kept as a separate "missing" category in Table 2; employment outcomes are computed only from non-missing components, so stores with a missing FTE component in either wave drop out of the corresponding regression (378 of 399 balanced stores).

3. Empirical design

The DiD estimand is

$$\delta = (NJ_{after} - NJ_{before}) - (PA_{after} - PA_{before})$$

We estimate it by ordinary least squares on first differences,

$$\Delta FTE_i = \alpha + \delta \cdot NJ_i + \varepsilon_i$$

which is numerically identical, in a balanced panel, to the two-way fixed-effects specification $FTE_{it} = \beta \cdot NJ_i + \gamma \cdot After_t + \delta \cdot (NJ_i \times After_t) + \varepsilon_{it}$. Standard errors are heteroskedasticity-robust (HC1). We then add controls successively (Table 4).

Identification. The identifying assumption is parallel trends: absent the wage increase, employment in NJ and PA would have evolved identically. With only two waves the assumption cannot be tested directly; we support it by (i) the regional proximity of the two states, (ii) a

compliance check showing the wage increase was actually implemented (Table 2), and (iii) robustness of the estimate to controls and sample choices (Table 6).

4. Core replication results and comparison with the original paper

Table 2 — compliance. Wave-1 starting wages are spread across $4.25–$5.06+ in both states. In wave 2, 85.5% of NJ stores report a starting wage in $5.01–$5.05, and the NJ mean wage rises from $4.61 to $5.08. Pennsylvania is unchanged (mean $4.63 → $4.62). The treatment was implemented as intended (Figure 1).

Table 3 — main DiD.

Outcome	NJ wave 1	NJ wave 2	NJ change	PA wave 1	PA wave 2	PA change	DiD	s.e.
FTE employment	20.60	21.36	+0.66	23.40	21.44	−2.06	**+2.72**	1.34
Headcount	29.99	30.64	+0.49	33.13	31.55	−1.84	**+2.33**	1.45

FTE employment rose in NJ and fell in PA, so the DiD estimate is positive: +2.72 FTE workers (t = 2.04). There is no evidence of a negative employment effect.

Table 4 — regression estimates (dependent variable: change in FTE).

Specification	NJ coefficient	s.e.	t	N
NJ only	+2.722	1.335	2.04	378
+ chain & ownership	+2.780	1.344	2.07	378
+ region dummies	+3.097	1.462	2.12	378
+ initial wage	+2.628	1.469	1.79	359

The estimate remains positive and of similar magnitude across specifications.

Comparison with the original paper. Card & Krueger's Table 3 reports a balanced-panel FTE DiD of **+2.75** (s.e. ≈ 1.36), with NJ employment rising about +0.6 FTE and PA falling about −2.2 FTE. Our estimate of **+2.72** (s.e. 1.34) is essentially identical. The small level differences (NJ change +0.66 vs. +0.59; PA change −2.06 vs. −2.16) arise from a minor sample difference: we use the `STATUS2 == 1` balanced panel with listwise deletion of missing employment

components (N = 378), whereas the original paper's balanced sample was constructed directly from the survey sheets. The headline conclusion — no employment decline, a point estimate that is positive — is reproduced exactly.

5. Independent extension: menu-price pass-through

Question. The minimum wage raised the cost of labour. Did restaurants pass any of that cost through to consumers via higher menu prices? The original paper discusses prices descriptively but does not estimate a formal DiD for them.

Design. We apply the same DiD design to the change in each of the three menu prices recorded in both waves (medium soda, small fries, entree), with chain and ownership controls:

$$\Delta Price_i = \alpha + \delta \cdot NJ_i + X_i'\gamma + \varepsilon_i$$

Results.

Menu item	NJ mean change	PA mean change	DiD	s.e.	t	N
Medium soda	+0.001	+0.008	−0.005	0.008	−0.61	379
Small fries	+0.017	+0.019	−0.001	0.008	−0.11	365
Entree	+0.043	−0.033	**+0.088**	0.043	2.06	374

Interpretation. Soda and fries show no detectable pass-through. The entree — the highest-priced item — rose by about 8.8 cents in NJ relative to PA (t = 2.06), roughly 6.5% of the ~$1.35 entree price. This is consistent with *partial, item-specific* pass-through: restaurants absorbed most of the wage increase (in line with the finding that employment did not fall) while adjusting a single higher-margin menu item. The result is economically modest relative to the $0.80 wage increase, indicating that most of the cost was absorbed through other margins (labour demand, hours, or margins).

6. Validation, limitations, and conclusion

Validation. (i) The compliance check (Table 2) confirms the treatment was binding and implemented. (ii) The main DiD is stable across controls (Table 4). (iii) Robustness checks (Table 6) confirm the result is not driven by sample choice: the estimate is +2.72 for FTE, +2.33 for headcount, and +2.42 when closed stores are treated as having zero wave-2 employment. Restricting to "compliers" — stores paying exactly the old $4.25 minimum before the increase —

strengthens the result to +3.64 (t = 2.04), consistent with a treatment-intensity story: the most-treated stores show the largest relative employment gain. The full-time/part-time split (+0.92 full-time, −1.56 part-time, both insignificant) suggests the relative gain was concentrated among full-time workers.

Limitations. (i) With only two waves, the parallel-trends assumption cannot be tested; the design relies on the regional comparison. (ii) Treatment is assigned by state, not at random, and the early 1990s recession may have hit the two states differently. (iii) Some NJ stores already paid ≥ $5.05 before the increase, attenuating the measured effect. (iv) Robust standard errors do not address spatial correlation across stores within a labour market. (v) Survey non-response and store closures are potential sources of selection, which we address but cannot fully rule out.

Conclusion. The replication reproduces Card & Krueger's central finding: New Jersey's minimum-wage increase did not reduce fast-food employment relative to Pennsylvania; the DiD point estimate is positive (+2.72 FTE, matching the paper's +2.75). The extension adds a new, testable margin — menu-price pass-through — and finds modest, item-specific pass-through concentrated in the entree, consistent with firms absorbing the cost largely through margins rather than employment or prices.

References

Card, David, and Alan B. Krueger. 1994. "Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania." *American Economic Review* 84(4): 772–793.